

4

Time Value of Money: Valuing Cash Flow Streams

LEARNING OBJECTIVES

- Value a series of many cash flows
- Value a perpetual series of regular cash flows called a *perpetuity*
- Value a common set of regular cash flows called an *annuity*
- Value both perpetuities and annuities when the cash flows grow at a constant rate
- Compute the number of periods, cash flow, or rate of return of a loan or investment
- Value cash flow streams with non-annual payments

notation

C	cash flow	N	date of the last cash flow in a stream of cash flows
C_n	cash flow at date n	P	initial principal or deposit, or equivalent present value
FV	future value	PV	present value
FV_n	future value on date n	r	interest rate or rate of return
g	growth rate		

As we have discussed, to evaluate a project a financial manager must compare its costs and benefits. In most cases, the cash flows in financial investments involve more than one future period. Thus, the financial manager is faced with the task of trading off a known up-front cost against a series of uncertain future benefits. We learned to value those costs and benefits by computing their cash values today—their present values.

In financial management, as well as your personal finances, you will need to evaluate series of cash flows occurring across time. In this chapter, we build on the tools we developed in Chapter 3 to value any series of cash flows. We will develop shortcuts for valuing *annuities*, *perpetuities*, and other special cases of assets with cash flows that follow regular patterns.

In Chapter 5, we will learn how interest rates are quoted and determined. Once we understand how interest rates are quoted, it will be straightforward to extend the tools of this chapter to cash flows that occur more frequently than once per year.

4.1 Valuing a Stream of Cash Flows

stream of cash flows
A series of cash flows lasting several periods.

We refer to a series of cash flows lasting several periods as a **stream of cash flows**. As with single cash flows, we can represent a stream of cash flows on a timeline. In this chapter, we will continue to use timelines and the rules of cash flow valuation introduced in Chapter 3 to organize and then solve financial problems.

Applying the Rules of Valuing Cash Flows to a Cash Flow Stream

Most investment opportunities have multiple cash flows that occur at different points in time. In Chapter 3, we learned the rules to value such cash flows:

Rule 1: Only values at the same point in time can be compared or combined.

Rule 2: To calculate a cash flow's future value, we must compound it using Eq. 3.1 from Chapter 3, which we repeat here.

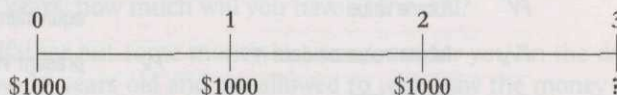
$$FV_n = C \times (1 + r)^n \quad (4.1)$$

Rule 3: To calculate the present value of a future cash flow, we must discount it using Eq. 3.2 from Chapter 3, which we repeat here.

$$PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n} \quad (4.2)$$

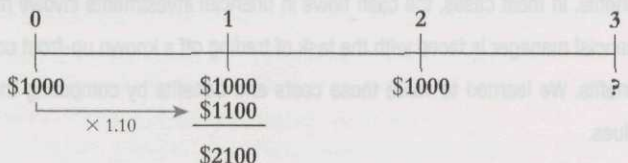
The rules of cash flow valuation allow us to compare and combine cash flows that occur at different points in time. Suppose we plan to save \$1000 today and \$1000 at the end of each of the next two years. If we earn a fixed 10% interest rate on our savings, how much will we have three years from today?

Again, we start with a timeline:

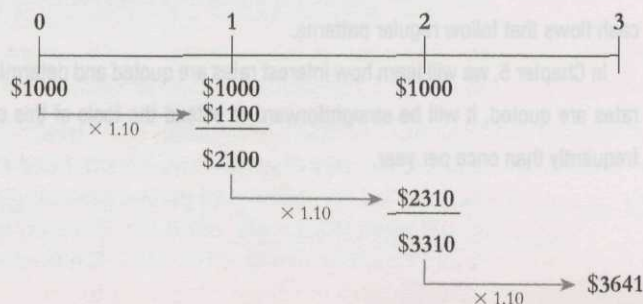


The timeline shows the three deposits we plan to make. We need to compute their value at the end of three years.

We can use the cash flow valuation rules in a number of ways to solve this problem. First, we can take the deposit at date 0 and move it forward to date 1. Because it is then in the same time period as the date 1 deposit, we can combine the two amounts to find out the total in the bank on date 1:

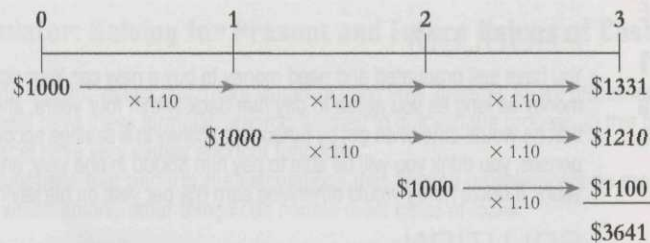


Using the first two rules, we find that our total savings on date 1 will be \$2100. Continuing in this fashion, we can solve the problem as follows:



The total amount we will have in the bank at the end of three years is \$3641. This amount is the future value of our \$1000 savings deposits.

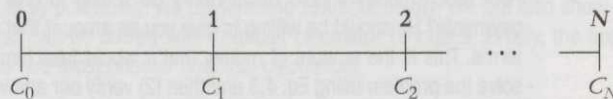
Another approach to the problem is to compute the future value in year 3 of each cash flow separately. Once all three amounts are in year 3 dollars, we can then combine them.



Both calculations give the same future value of \$3641. As long as we follow the rules, we get the same result. The order in which we apply the rules does not matter. The calculation we choose depends on which is more convenient for the problem at hand.

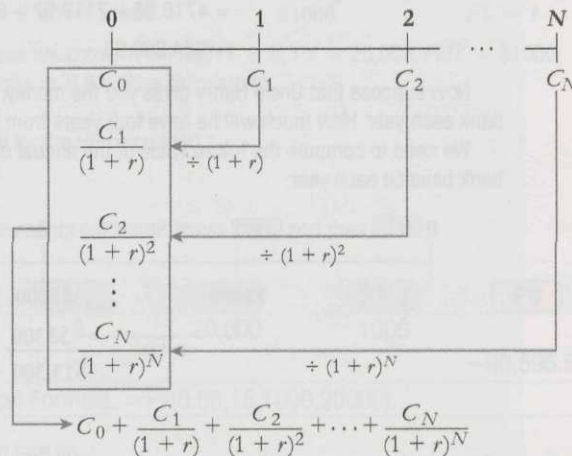
Now we formalize this approach by deriving a general formula for valuing a stream of cash flows.

Consider a stream of cash flows: C_0 at date 0, C_1 at date 1, and so on, up to C_N at date N . We represent this cash flow stream on a timeline as follows:



Using the rules of cash flow valuation, we compute the present value of this cash flow stream in two steps. First, we compute the present value of each individual cash flow. Then, once the cash flows are in common units of dollars today, we can combine them.

For a given interest rate r , we represent this process on the timeline as follows:



This timeline provides the general formula for the present value of a cash flow stream:

$$PV = C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N}{(1+r)^N} \quad (4.3)$$

That is, the present value of the cash flow stream is the sum of the present values of each cash flow. Recall from Chapter 3 that we defined the present value as the dollar amount you would need to invest today to produce the single cash flow in the future. The same idea holds in this context. The present value is the amount you need to invest today to generate the cash flows stream $C_0, C_1, \dots, C_N$. That is, receiving those cash flows is equivalent to having their present value in the bank today.

Example 4.1 illustrates that if you want to compute the future value of a stream of cash flows, you can do it directly (the first approach), or you can first compute the present value and then move it to the future (the second approach). As always, we use Eq. 4.1 to

PERSONAL FINANCE

EXAMPLE 4.1

Present Value of a Stream of Cash Flows

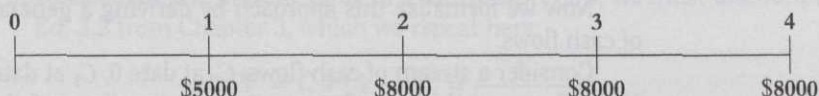
PROBLEM

You have just graduated and need money to buy a new car. Your rich Uncle Henry has offered to lend you the money as long as you agree to pay him back within four years, and you offer to pay him the rate of interest that he would otherwise get by putting his money in a savings account. Based on your earnings and living expenses, you think you will be able to pay him \$5000 in one year, and then \$8000 each year for the next three years. If Uncle Henry would otherwise earn 6% per year on his savings, how much can you borrow from him?

SOLUTION

PLAN

The cash flows you can promise Uncle Henry are as follows:



How much money should Uncle Henry be willing to give you today in return for your promise of these payments? He should be willing to give you an amount that is equivalent to these payments in present value terms. This is the amount of money that it would take him to produce these same cash flows. We will (1) solve the problem using Eq. 4.3 and then (2) verify our answer by calculating the future value of this amount.

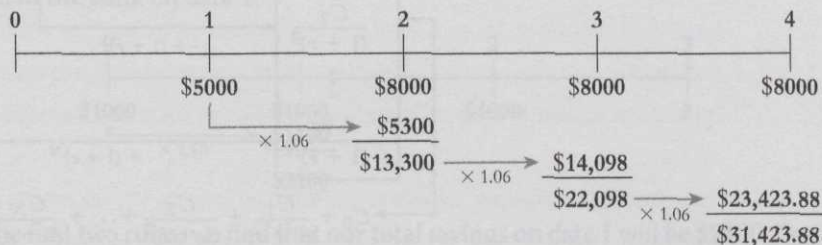
EXECUTE

1. We can calculate the PV as follows:

$$\begin{aligned} PV &= \frac{5000}{1.06} + \frac{8000}{1.06^2} + \frac{8000}{1.06^3} + \frac{8000}{1.06^4} \\ &= 4716.98 + 7119.97 + 6716.95 + 6336.75 \\ &= \$24,890.65 \end{aligned}$$

Now suppose that Uncle Henry gives you the money, and then deposits your payments to him in the bank each year. How much will he have four years from now?

We need to compute the future value of the annual deposits. One way to do this is to compute the bank balance each year:



2. To verify our answer, suppose your uncle kept his \$24,890.65 in the bank today earning 6% interest. In four years he would have:

$$FV = \$24,890.65 \times (1.06)^4 = \$31,423.87 \text{ in 4 years}$$

We get the same answer both ways (within a penny, which is due to rounding).

EVALUATE

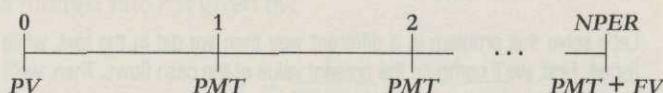
Thus, Uncle Henry should be willing to lend you \$24,890.65 in exchange for your promised payments. This amount is less than the total you will pay him (\$5000 + \$8000 + \$8000 + \$8000 = \$29,000) due to the time value of money.

calculate the future value of any present value. Because we obey the rules of valuing cash flows in both cases, we get the same result.

Using a Financial Calculator: Solving for Present and Future Values of Cash Flow Streams

So far, we have used formulas to compute present values and future values of cash flow streams. As we discussed at the end of Chapter 3, both financial calculators and spreadsheets have these formulas preprogrammed to quicken the process. In this box, we focus on financial calculators, but spreadsheets such as Excel have very similar shortcut functions.

Financial calculators have a set of functions that perform the calculations that finance professionals do most often. These functions are all based on the following timeline, which among other things can handle most types of loans:

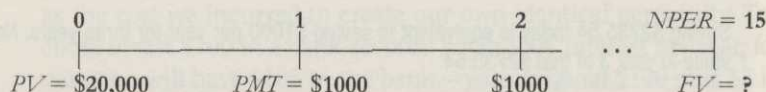


There are a total of five variables: number of periods (N or $NPER$), present value (PV), cash flow or "payment" (PMT), future value (FV), and the interest rate, denoted I/Y . Each function takes four of these variables as inputs and returns the value of the fifth one that ensures that the sum of the present value of the cash flows is zero.

By setting the recurring payments equal to 0, you could compute present and future values of single cash flows such as we have done above using Eqs. 4.2 and 4.1. In the examples shown in Sections 4.2 through 4.4, we will calculate cash flows using the **PMT** button. The best way to learn to use a financial calculator is by practicing. We present one example below. We will also show the calculator buttons for any additional examples in this chapter that can be solved with financial calculator functions. Finally, the appendix to this chapter contains step-by-step instructions for using the two most popular financial calculators.

Example

Suppose you plan to invest \$20,000 in an account paying 8% interest. You will invest an additional \$1000 at the end of each year for 15 years. How much will you have in the account in 15 years? We represent this problem with the following timeline:



To compute the solution, we enter the four variables we know, $N = 15$, $I/Y = 8$, $PV = 20,000$, $PMT = 1000$, and solve for the one we want to determine: FV . Specifically, for the HP-10bII+ or TI BAII Plus calculators:

1. Enter 15 and press the **N** button.
2. Enter 8 and press the **I/Y** button (**I/YR** for the HP calculator).
3. Enter 20,000 and press the **PV** button.
4. Enter \$1000 and press the **PMT** button.
5. Press the **FV** button (for the Texas Instruments calculator, press **CPT** and then **FV**).

	N	I/Y	PV	PMT	FV
Given:	15	8	20,000	1000	
Solve for:					-90,595.50
Excel Formula: =FV(0.08,15,1000,20000)					

The calculator then shows a future value of $-\$90,595.50$.

Note that we entered PV and PMT as positive numbers (the amounts we are putting into the bank), and FV is shown as a negative number (the amount we can take out of the bank). Calculators and spreadsheets balance the PV 's of the cash flows, so some of the inputs must be negative and some must be positive. It is critical to be consistent: in this case, all money that we contributed was entered positively, and the money we got out at the end was negative. Alternatively, it would have been fine to enter all money we contributed with a negative sign (the PV and the PMT), with the result being that the FV would be presented as a positive flow. You will see more examples of getting the sign of the cash flows correct throughout the chapter.

Excel has the same functions, but it calls " N ," " $NPER$ " and " I/Y ," " $RATE$." **Also, it is important to note that you enter an interest rate of 8% as "8" in a financial calculator, but as "0.08" in Excel.**

At the end of this chapter on page 130, we highlight the features of our interactive annuity calculator that is available in MyLab Finance and is integrated into the eTextbook.